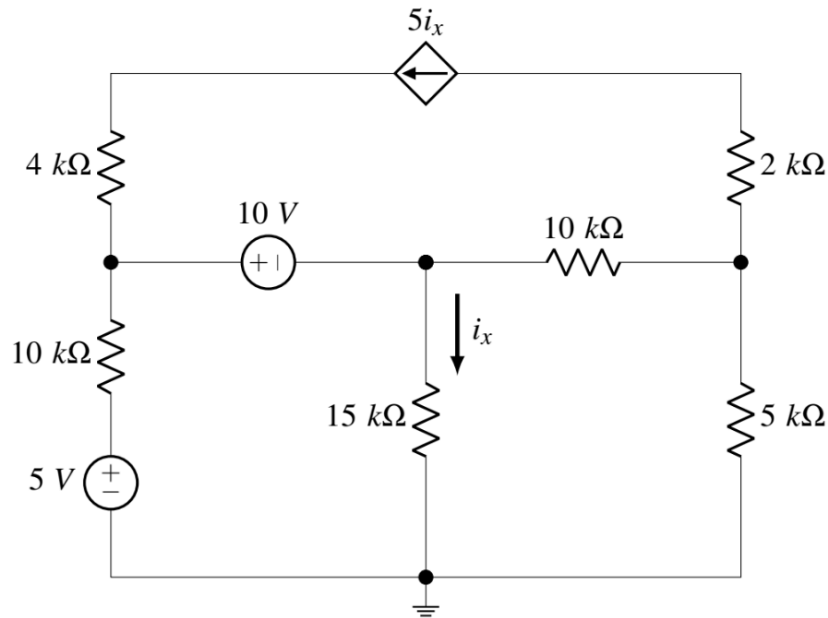


Assignment - 01

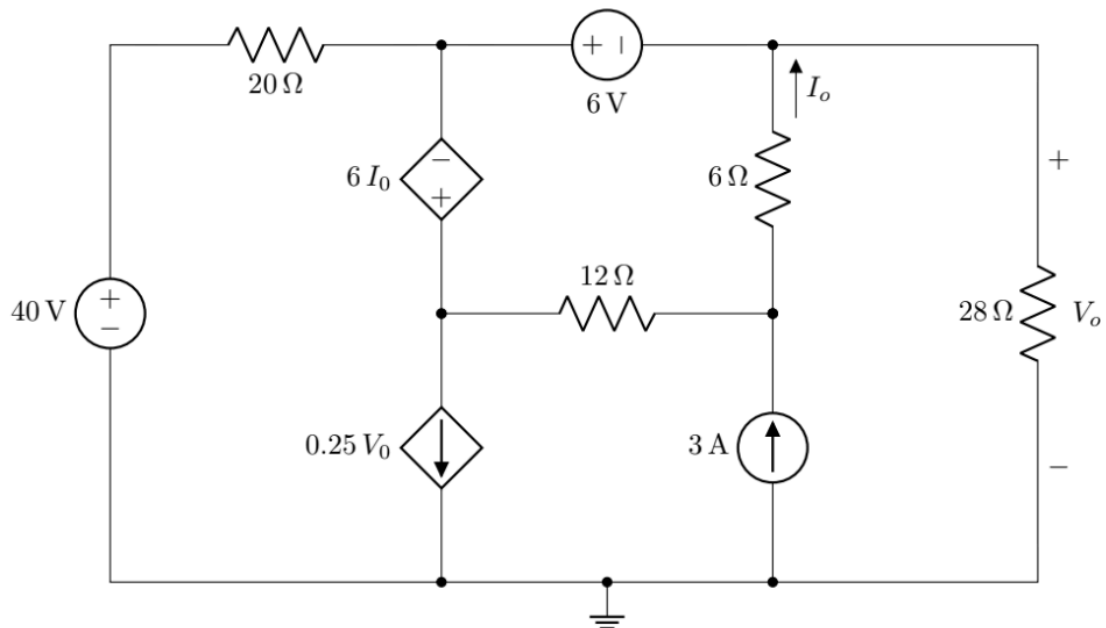
Course: CSE250

Faculty: Sheikh Iftekhar Ahmed [SEF]

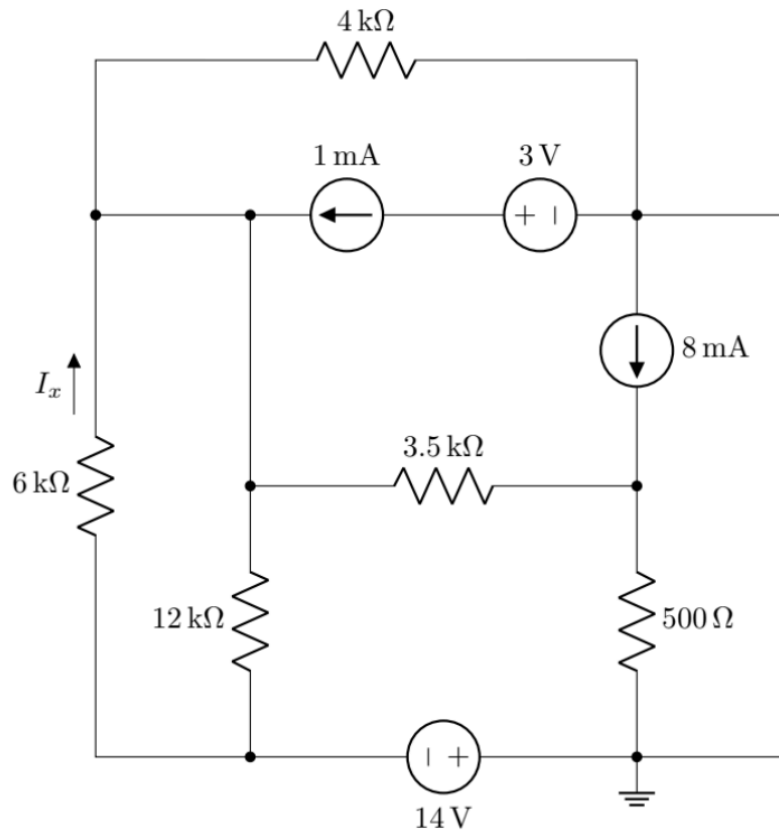
- 1) Apply Nodal Analysis in the following circuit to find all the Node voltages.
Again, solve the same circuit using Mesh Analysis and find all Mesh currents.



- 2) Apply Nodal or Mesh Analysis (Either one as per your choice, not both) to find the power supplied or absorbed by the dependent voltage source.



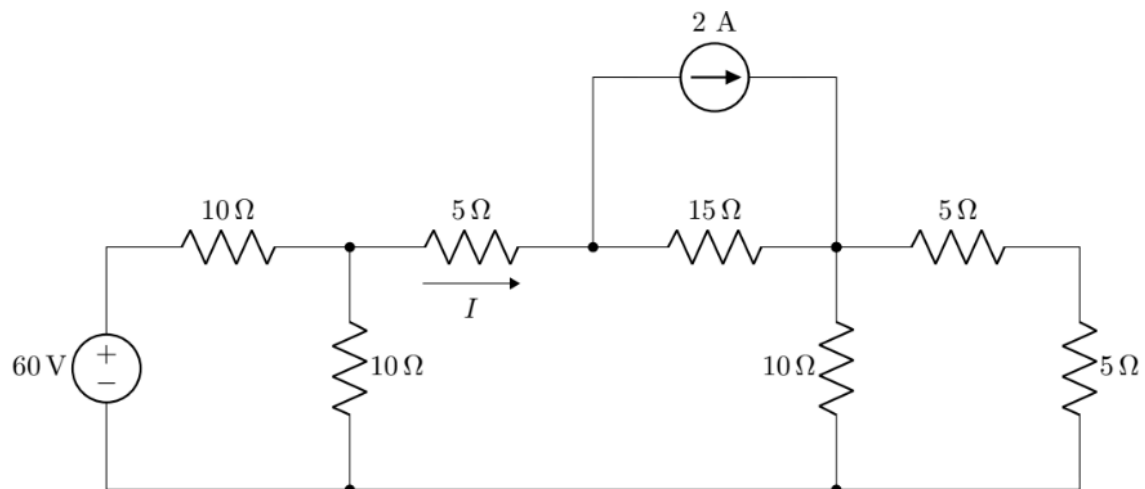
3) Analyze the following circuit and answer the questions that follow:



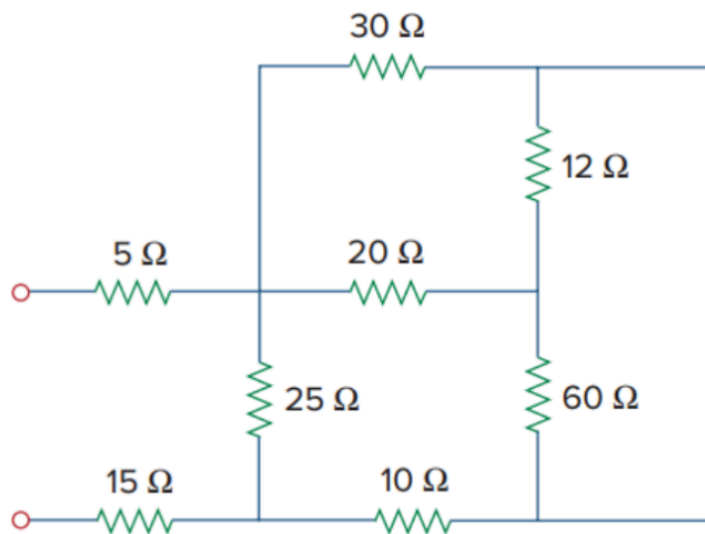
(a) You can use either Nodal or Mesh Analysis. Which one would be easier for this circuit?

(b) Using your selection method in part (a), find the current ' I_x '.

4) Find the current ' I ' in the circuit using the Superposition Principle.



- 5) Find the equivalent resistance of the following circuit, across the two red circular nodes.



- 6) Apply Mesh Analysis to find the voltage across the 3 A Current source.

